PULSE-AI v1: An Interpretable, Validated ECG Marker Engine for Atrial and Ventricular Assessment

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***Abstract*—Damage to the left atrium begins years before atrial fibrillation (AF) and can be seen on a normal electrocardiogram (ECG). Doctors read it from a few named measurements of the P wave, such as its duration and the P-terminal force in lead V1 (PTFV1). Most automated ECG systems do not report these measurements: they either detect AF once it is already present, or they give a risk score that cannot be checked by hand. We present PULSE-AI v1, a signal-processing pipeline that measures a set of named markers from a standard 10-second 12-lead ECG: atrial markers (P duration, PTFV1, P axis, interatrial block) and ventricular markers (QRS duration, QT and corrected QT, T-peak to T-end interval, ST level). Each P, QRS and T wave is located with a wavelet method, and the timing error was measured against cardiologist annotations on held-out data: the P-duration error was +5.6 ms on average, and three of five boundaries met the CSE tolerances. We then compared 259 ECGs labelled left atrial enlargement (LAE) with 516 age- and sex-matched controls from the public PTB-XL database, using an analysis plan fixed before the data were examined. P duration did not differ between the groups (difference +2.0 ms, 95% confidence interval −0.9 to +5.0 ms), and the usual 120 ms threshold marked more than nine in ten ECGs in both groups as abnormal. PTFV1 differed strongly; because the LAE label is partly read from the P wave in V1, this confirms the measurement rather than discovering new disease. Ventricular markers outside the LAE definition also differed, and kept their direction after excluding left ventricular hypertrophy and bundle branch block. A random forest separated the two groups with an area under the ROC curve of 0.830, and 0.746 when every P-wave marker was removed. The results show that these markers can be measured automatically with known error, and that in older patients PTFV1 is more useful than P duration at a fixed 120 ms threshold.**

***Index Terms*—electrocardiogram, P wave, PTFV1, wavelet delineation, atrial cardiopathy, interpretable machine learning, PTB-XL**

I. Introduction

Atrial fibrillation (AF) raises the risk of stroke about five times [1]. It is often silent and comes and goes, so many patients have a stroke before AF has ever been recorded. AF does not appear suddenly in a healthy heart. The left atrium first becomes enlarged and scarred, a condition called atrial cardiopathy, and this slows the electrical signal that travels through it (Fig. 1). The slower signal changes the P wave of the ECG while the rhythm is still normal (sinus rhythm). In the BAYES registry, people with advanced interatrial block (IAB), a long and notched P wave, developed AF 3.31 times as often (95% CI 1.87–5.86) as people without it [2]. The P-terminal force in V1 (PTFV1) [3] and an abnormal P axis [4] show similar associations. Each of these markers has a published definition and threshold [5], [6].

Automated ECG analysis has mostly followed two other paths. Smartwatches and patch monitors detect AF while it is happening. Deep neural networks trained on raw ECGs can flag patients who had AF from a sinus-rhythm tracing [7], but the reasons for their output cannot be read by the doctor. A third path has received less attention: measuring the named markers automatically and reporting how accurate each measurement is, so that a doctor can read the number and a study can select patients by it.

This paper describes the first version of our pipeline, PULSE-AI v1, and makes four contributions:

1) A pipeline that measures named atrial and ventricular ECG markers, in which every threshold is linked to its published source.
2) A test of the wave-boundary detection against cardiologist annotations on held-out and external data, with the error given in milliseconds.
3) A case–control comparison on PTB-XL whose analysis plan was fixed in advance, with a quality check that cannot see which group a record belongs to.
4) A classifier analysis that separates the information carried by P-wave markers from that carried by ventricular markers.

II. Related Work

Wave-boundary detection. The wavelet method of Martínez et al. [8] and the multilead rule of Laguna et al. [9] are standard methods. Both were tested on the QT Database (QTDB) [10] against the tolerances of the Common Standards for Electrocardiography (CSE) project [11]. The Lobachevsky University Database (LUDB) gives expert boundaries for all 12 leads [12].

P-wave reference values. Normal values of P duration depend on how it is measured. Measuring across all leads at once gives longer durations than reading one lead [13]. With a lead-independent algorithm, the median P duration of healthy men aged 60–89 is about 120 ms [14]. P duration

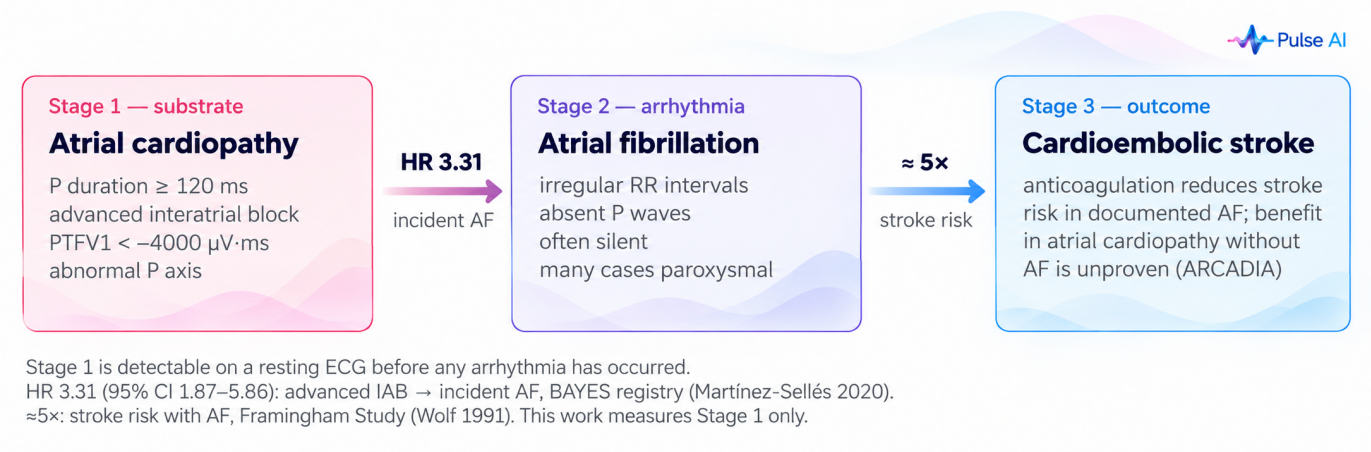


Fig. 1. The three-stage chain from atrial damage to stroke. PULSE-AI v1 measures Stage 1 only and makes no treatment claim.

TABLE I
ABBREVIATIONS

AF	atrial fibrillation	LUDB	Lobachevsky Univ. Database
AUROC	area under ROC curve	MVP	morphology-voltage-P score
AUPRC	area under prec.–recall curve	OLS	ordinary least squares
BH	Benjamini–Hochberg	PPG	photoplethysmography
CI	confidence interval	PTFV1	P-terminal force in V1
CSE	Common Standards for ECG	QC	quality control
ECG	electrocardiogram	QTc	heart-rate-corrected QT
HR	heart rate	QTDB	QT Database
IAB	interatrial block	RMSSD	root mean square of successive RR differences
IQR	interquartile range		
LAE	left atrial enlargement	SMD	standardised mean diff.
LAO	left atrial overload	Tpe	T-peak to T-end interval

increases with age [15], and in a large primary-care cohort both short and long P waves were linked to AF [16]. The MVP score combines P-wave shape, voltage and duration into one number [17].

Learned ECG models. PTB-XL [18] is the standard public data set for 12-lead ECG classification. Deep models reach high accuracy on it, but they do not output quantities a doctor can check [7].

III. Methods

Table I lists the abbreviations used in this paper.

A. Data and Cohort

PTB-XL v1.0.3 [18] contains 21,799 ECGs of 10 s from 18,869 patients; we used the files sampled at 500 Hz. Cases were sinus-rhythm ECGs carrying the diagnostic label LAO/LAE (left atrial overload or enlargement). Controls were sinus-rhythm ECGs of patients who never carry that label. We used one ECG per patient and matched two controls to each case, exactly on sex and within ±5 years of age, keeping patients aged 40 or over. Of 332 eligible cases, 5 were younger than 40; every remaining case was matched, giving 327 cases and 654 controls with a mean age of 68.3 years. Left ventricular hypertrophy (LVH), bundle branch block, Wolff–Parkinson–White pattern and paced rhythm were not exclusion criteria; their effect was tested afterwards (Section IV-E). A second design (Design B) used the sinus-rhythm ECGs of patients who had AF recorded in another PTB-XL ECG (118 cases, 236 controls). The sinus ECG could lie before or after the AF ECG (51 and 67 of the 118 cases), so Design B tests association with an AF history, not prediction of future AF. All data splits were grouped by patient, so no patient appears in both training and test data.

B. Filtering and Median Beat

Each lead $x_\ell[n]$, $\ell = 1, \ldots, 12$, was filtered with a 50 Hz notch ($Q = 30$) and a 0.5–40 Hz fourth-order Butterworth band-pass $h[n]$. PTB-XL was recorded on 50 Hz mains; the band-pass alone already attenuates 50 Hz by 17.9 dB, and the raw records showed a median 50 Hz peak only 0.5 dB above the neighbouring noise floor, so the notch is a safeguard for the few noisy records (6 with a peak above 10 dB). The filter was applied forward and backward,

$$y_\ell[n] = h[-n] * \big(h[n] * x_\ell[n]\big), \qquad Y_\ell(e^{j\omega}) = |H(e^{j\omega})|^2 \, X_\ell(e^{j\omega}), \tag{1}$$

which gives zero phase shift, so wave boundaries are not moved in time. R peaks $r_1, \ldots, r_K$ were found on all leads together, and a median beat was formed for each lead,

$$\bar{y}_\ell[n] = \underset{k=1,\ldots,K}{\text{median}} \; y_\ell[r_k + n], \qquad -0.40 f_s \le n \le 0.50 f_s, \tag{2}$$

with $f_s = 500$ Hz. Noise varies from beat to beat and is suppressed by the median, while the P wave, which appears at the same position in every sinus beat, is kept. Premature or ectopic beats were not removed before averaging; the median tolerates them only while they are a minority of the beats, and mean HR, RMSSD and the RR used for QTc include every interval.

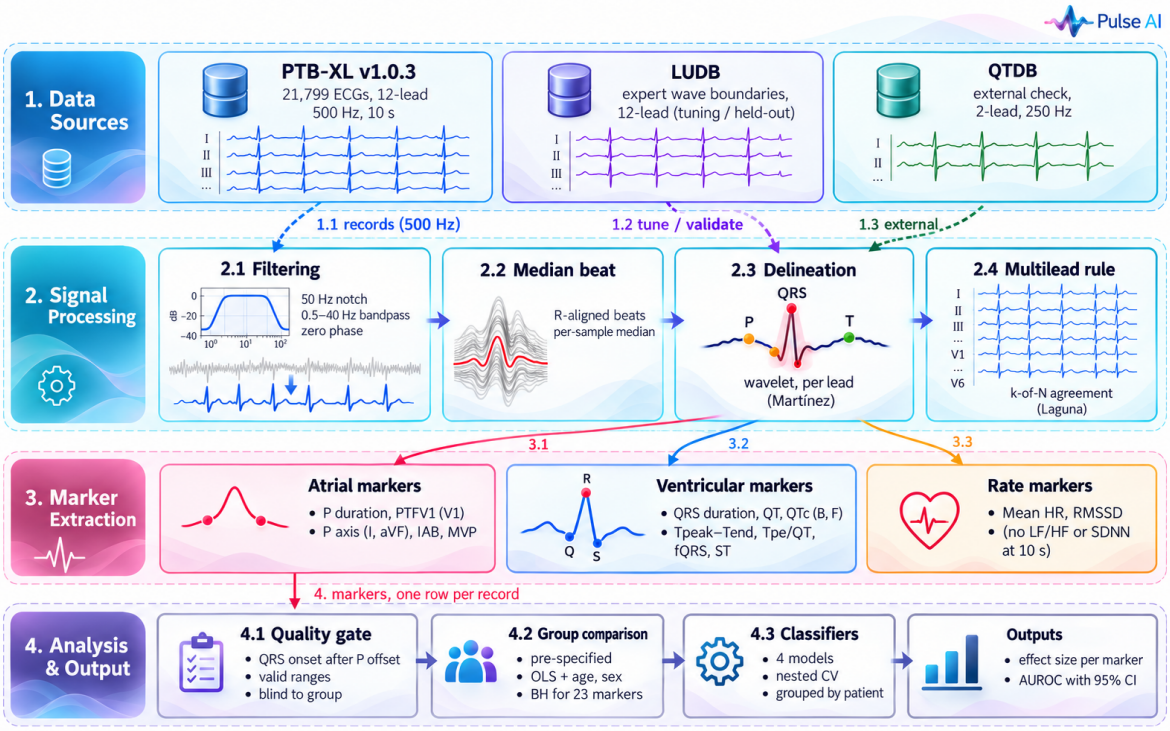


Fig. 2. PULSE-AI v1 pipeline. Numbers on the arrows give the order of operations. Dashed arrows: data used only to tune and test the boundary detector. Red: atrial markers; blue: ventricular markers; orange: rate markers. Waveforms and icons are schematic.

C. Wave-Boundary Detection and Its Validation

Each median beat was analysed with the dyadic wavelet transform of Martínez et al. [8],

$$W_{2^k}\bar{y}[n] = \sum_m \bar{y}[m]\,\psi_{2^k}[n-m], \qquad k = 1,\dots,5, \tag{3}$$

where ψ is the derivative of a quadratic spline, so each scale behaves as a smoothed derivative. A wave is accepted when $|W_{2^k}\bar{y}|$ exceeds a threshold proportional to its root-mean-square value; its peak lies at a zero crossing of $W_{2^k}\bar{y}$ and its onset and offset where the modulus falls below a fixed fraction of the neighbouring maximum. The boundaries found in the 12 leads were combined with the k-of-N rule of Laguna et al. [9]: a global onset is the earliest time at which at least k leads agree within δ ms ($k = 3$ for QRS and T; $k = 2$, $\delta = 10$ ms for P). Our changes to the published method (search windows, a 0.02 mV minimum P-wave height and a 100 ms P-wave slope span) are documented in the code. LUDB [12] was split by record into a tuning half and a held-out half. Every parameter was set on the tuning half only, and all accuracy figures are from held-out sinus-rhythm records. QTDB served as an external check with no tuning: it was processed at its native 250 Hz, the rate at which the wavelet scales are defined [8], using both leads and the first cardiologist's annotation (q1c); 104 of 105 records (3,178 beats) were scored, one having no annotated QRS. Synthetic ECGs built from half-sine waves with exactly known boundaries were used only for unit tests and to measure systematic bias of the detector, never in any group analysis.

D. Markers

Table II lists the markers. With t_{on} and t_{off} the global boundaries, the main markers are

$$\text{P duration} = t^P_{\text{off}} - t^P_{\text{on}}, \tag{4}$$

$$\text{PTFV1} = -A_{\text{neg}} \times D_{\text{neg}} \quad [\mu\text{V}\cdot\text{ms}], \tag{5}$$

$$\theta_P = \operatorname{atan2}\!\left(\tfrac{2}{\sqrt{3}}\,S_{\text{aVF}},\; S_{\text{I}}\right), \tag{6}$$

$$\text{QTc}_{\text{B}} = \frac{\text{QT}}{\sqrt{\text{RR}}}, \qquad \text{QTc}_{\text{F}} = \frac{\text{QT}}{\sqrt[3]{\text{RR}}}, \tag{7}$$

$$\text{RMSSD} = \sqrt{\tfrac{1}{N-1}\textstyle\sum_{i=1}^{N-1}(\text{RR}_{i+1} - \text{RR}_i)^2}, \tag{8}$$

where A_{neg} and D_{neg} are the depth and duration of the terminal negative part of the P wave in V1 (measured with V1's own boundaries), S_{I} and S_{aVF} are the P-wave areas in leads I and aVF (the factor $2/\sqrt{3}$ follows from the hexaxial lead geometry), and RR is the mean beat interval in seconds. Fragmented QRS (fQRS) is read from a separate 0.5–100 Hz median beat, because the 40 Hz low-pass would smooth the notches it looks for. Tpe runs from the maximum of the spatial magnitude $\sqrt{\sum_\ell \bar{y}_\ell^2[n]}$ in the T wave to the T offset. IAB is reported only descriptively. QTc is analysed as a continuous value; the single 460 ms threshold in Table II is the female limit of [19] applied to both sexes and is not used in any test. Heart-rate-variability measures that need at least 300 s of signal (SDNN, LF/HF, sample entropy, DFA) [20] are not computed on 10-s records. If a record lacks a required lead, the marker is returned as "not applicable" with the reason, so a single-lead recording skips PTFV1 automatically.

TABLE II
MARKERS, LEADS USED AND PUBLISHED THRESHOLDS

Marker	Leads	Abnormal	Ref.
Atrial			
P duration	all 12	$\geq$120 ms	[5]
PTFV1	V1	$\leq -4000\,\mu$V·ms	[3]
P axis	I, aVF	outside 0–75°	[4]
IAB class	II, III, aVF	P $\geq$120 ms ($\pm$ biphasic)	[5]
MVP score	12	$\geq$5 of 6	[17]
Ventricular			
QRS duration	all 12	$\geq$120 ms	[21]
QTc (B, F)	all 12	$\geq$460 ms	[19], [22], [23]
Tpe, Tpe/QT	all 12	Tpe/QT >0.25	[24]
fQRS	territories	$\geq$2 leads	[25]
ST at J+60 ms	each lead	—	—
Rate			
Mean HR, RMSSD	12	—	[20]

E. Quality Control

A record was excluded if a required marker was missing, if HR, QRS, QT or P duration was outside physiological limits, or if the detected QRS onset came before the P offset on the median beat. The last case cannot occur in a real heart and shows that the boundaries were placed wrongly. The quality check receives only the measurements of a record, never its group label (this is enforced by a unit test), and every exclusion is counted under a named rule.

F. Statistical Analysis Fixed in Advance

The analysis plan was written and saved before any group difference was computed. The primary endpoint was P duration, treated as a continuous value and modelled by ordinary least squares,

$$m_i = \beta_0 + \beta_1\,\text{case}_i + \beta_2\,\text{age}_i + \beta_3\,\text{sex}_i + \varepsilon_i, \tag{9}$$

where β_1 is the adjusted difference between cases and controls. Matched sets broken by the quality check were analysed unmatched, with age and sex in the model. Effect sizes were Hedges' g,

$$g = \Big(1 - \frac{3}{4(n_1+n_2)-9}\Big)\frac{\bar{m}_1 - \bar{m}_2}{s_p}, \tag{10}$$

with s_p the pooled standard deviation, and Cliff's δ. The 23 secondary markers used the same model, with the false discovery rate held at 5% by the Benjamini–Hochberg (BH) procedure. Threshold calls (IAB, abnormal PTFV1, high MVP) were reported as percentages only. LAE labels in PTB-XL are given partly from P-wave criteria, so a difference in P-wave markers confirms that our measurement agrees with the label rather than discovering something new. Markers outside the LAE definition do not have this problem. Sensitivity analyses used the complete matched sets (paired test) and Design B. Checks added after the plan, and labelled post hoc wherever they appear, were: reports confirmed by a cardiologist only; exclusion of, or adjustment for, LVH and conduction disease; a comparison of included and excluded records; and a validation of PTFV1 against expert V1 boundaries.

G. Classifiers (Exploratory)

This part was added after the plan and is reported as exploratory. Four models were compared: L2-regularised logistic regression, a decision tree of depth 2–5, a random forest and histogram gradient boosting. Nested cross-validation was used: an outer 5-fold split, stratified and grouped by patient, measured performance, and an inner 3-fold split chose the hyperparameters and the decision threshold (Youden's J). Missing-value filling and scaling were fitted inside each training fold only. Feature set A held all markers plus age and sex. Set B removed every marker derived from the P wave (P duration, PTFV1, P axis, IAB, MVP), so its result cannot come from the information used to assign the label. Discrimination was measured by

$$\text{AUROC} = P\big(\hat{s}_{\text{case}} > \hat{s}_{\text{control}}\big), \tag{11}$$

the probability that a random case scores above a random control, and by the AUPRC, both with 1000-sample bootstrap CIs and beside a model that always predicts "control". The whole nested procedure was repeated on 200 random shuffles of the labels, giving $p = (1 + \#\{\text{AUROC}_{\text{perm}} \geq \text{AUROC}\})/(1+200)$. Deep neural networks were not used on purpose: with $n = 775$ they cannot be trained reliably and would not give readable outputs.

H. Software

Each marker is declared once in code together with its leads, units, normal range, threshold and reference. The quality check, Table II and the device-capability map are generated from these declarations, and every number in this paper is produced by a script from saved results. The code, analysis plan and result files are available at https://github.com/Huzaimkhani/pulse-ai-v1.

IV. Results

A. Cohort

Of 981 matched ECGs, 206 (21.0%) were excluded (Fig. 3). The main reason was a QRS onset placed before the P offset (152 ECGs; 12.2% of cases and 17.1% of controls). 259 cases and 516 controls remained, and they were still balanced in age (68.3 $\pm$ 11.9 vs. 68.2 $\pm$ 12.1 years, standardised mean difference 0.006) and sex (58.7% vs. 60.1% male, -0.028). The exclusion rate was the same in both groups (20.8% vs. 21.1%, $p = 0.91$; post hoc), and excluded ECGs did not differ from included ones in age, sex, HR or number of beats. They did have longer P, QRS and QTc values, as expected, since these are the values the quality rules test.

B. Accuracy of Wave-Boundary Detection

On 87 held-out LUDB sinus records (711 scored P waves, P-wave detection 99.9%), the within-record standard deviations of the boundary error were 15.2, 12.2, 12.0, 6.4 and 24.5 ms for P onset, P offset, QRS onset, QRS offset and T offset (Table III). P offset, QRS offset and T offset met the CSE tolerances; P onset and QRS onset did not. For each record, the

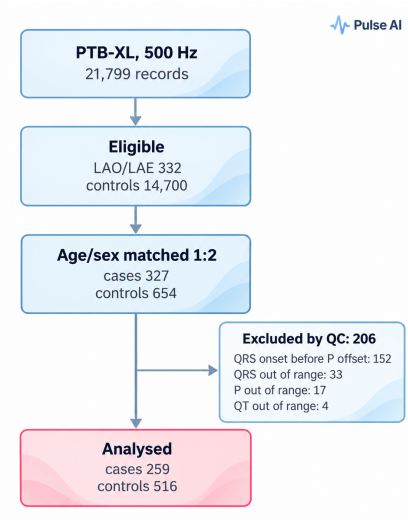


Fig. 3. Cohort flow. Each excluded ECG is counted under the first rule it failed.

TABLE III
BOUNDARY ERROR ON HELD-OUT LUDB SINUS RECORDS (MS)

Boundary	Mean	SD_{within}	CSE limit	Ref. [8]
P onset	−4.6	15.2	10.2	14.8
P offset	+3.1	12.2	12.7	12.8
QRS onset	−7.1	12.0	6.5	7.7
QRS offset	+0.2	6.4	11.6	8.7
T offset	+16.6	24.5	30.6	18.1

SD_{within}: mean of the per-record standard deviations, the statistic used in [8]. Their values (QTDB, best lead per point) are shown for orientation only.

error in P duration was +5.6 ± 23.4 ms (mean absolute error 16.3 ms), and +3.2 ± 19.6 ms on QTDB. Beat by beat, 95% of the errors lay between −53 and +68 ms (Fig. 4, limits of agreement = bias ±1.96 SD). Agreement on the 120 ms yes/no decision was low (Cohen's $\kappa = 0.26$), partly because the expert reference itself placed 66% of LUDB sinus records at or above 120 ms. On synthetic ECGs the measured P duration followed $0.782 \times$ true + 27.9 ms, so true differences between groups are slightly reduced in our measurements.

PTFV1 uses V1's own boundaries, which were checked separately (post hoc) against the expert V1 marks on the same 86 held-out records. V1 P onset agreed (−2.3 ± 42.2 ms), but our V1 P offset was late by +37.1 ± 40.4 ms. This lengthens the terminal negative part (+14.3 ms) without changing its depth (+1.2 µV). PTFV1 computed with our boundaries and with the expert boundaries on the same signal agreed well as a continuous value (ICC 0.93, Spearman $\rho = 0.92$; bias −605 μV·ms), and the abnormal call agreed with $\kappa = 0.78$. All disagreements went one way: 7 records called abnormal by us were normal with expert boundaries, and 0 the other way (28% vs. 21% abnormal). PTFV1 is therefore reliable as a continuous measure, but its threshold call over-reports.

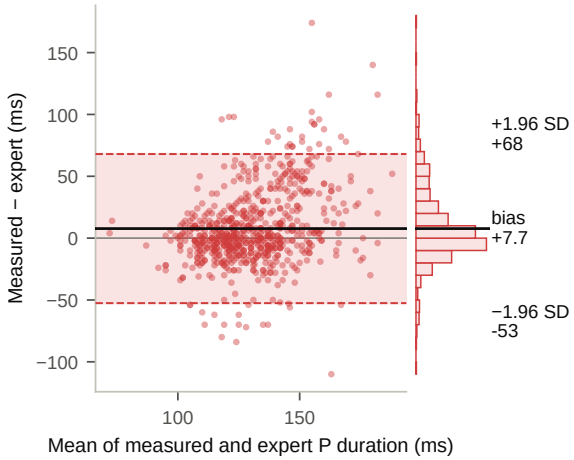


Fig. 4. Bland–Altman plot of measured P duration against expert annotation, held-out LUDB sinus beats. Shaded band: 95% limits of agreement (−53 to +68 ms).

C. Calibration on PTB-XL

Compared with normal ranges taken from single-lead reading (P 80–120 ms, QRS 70–110 ms), the cohort medians of P duration (143 ms) and QRS duration (112 ms) looked abnormal. Compared with normal values measured the same way and in the same age range [14], the remaining offsets were 18–28 ms for P and 12–17 ms for QRS, and similar in men and women. A constant offset moves cases and controls by the same amount. It makes absolute values and fixed thresholds unreliable, but it does not affect the comparison between groups. PR, QT and HR were within published ranges.

D. Case–Control Comparison

The primary endpoint showed no difference (Fig. 5). Median P duration was 146 ms in cases and 141 ms in controls; the adjusted difference was +2.0 ms (95% CI −0.9 to +5.0; $p = 0.17$; $g = +0.10$). The 120 ms threshold marked 93.4% of cases and 93.6% of controls as IAB, and the high-risk MVP group held 45.9% and 45.0%.

PTFV1 behaved differently. It was more negative in cases by −3627 μV·ms (95% CI −4237 to −3017; g =-0.88; q<0.001), and 81.8% of cases against 50.0% of controls passed the abnormal threshold. These percentages are descriptive only: they include the over-calling shown above, and PTB-XL controls also had deeper terminal negative P waves than LUDB records (median 50 vs. 26 µV), which boundary error cannot cause. Of the 23 secondary markers, 20 differed at $q < 0.05$ (Fig. 6). Among markers outside the LAE definition, cases had a longer QRS (+14.2 ms, $g = +0.54$), longer QTc (Fridericia, +32.8 ms, $g = +0.62$), longer Tpe (+17.7 ms), higher HR (+4.9 bpm), ST depression in I, aVL, V5 and V6 (g =-0.59 in V6) and ST elevation in aVR and V1–V3.

E. Sensitivity Analyses

In the 159 complete matched sets the paired difference in P duration was +2.5 ms (−1.1 to +6.2; $p = 0.17$). Using only

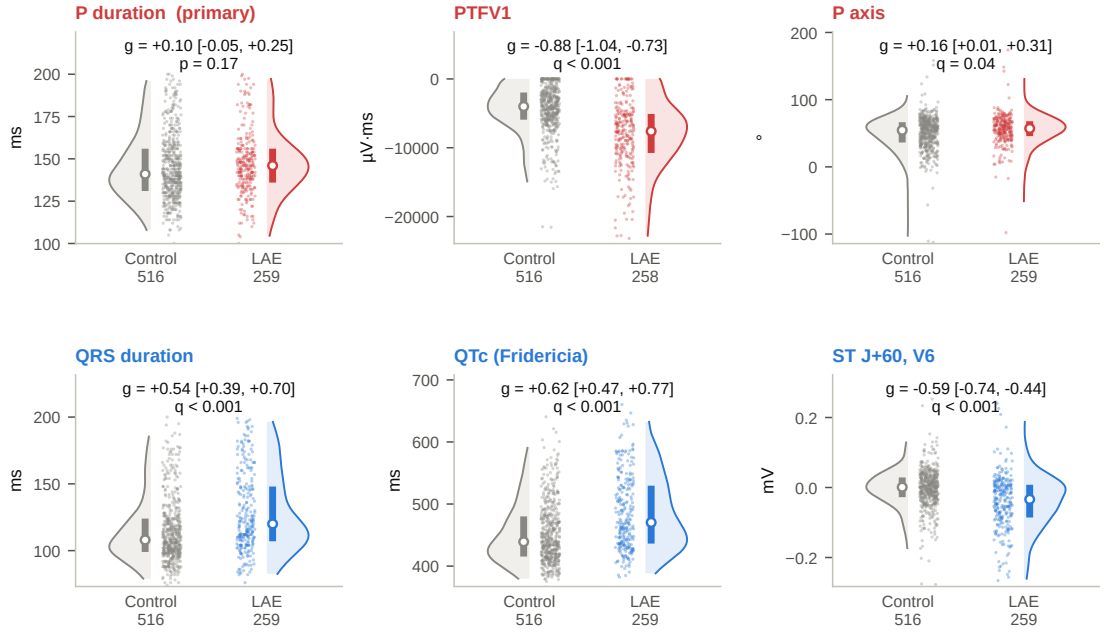


Fig. 5. Distribution of three atrial markers (top) and three ventricular markers (bottom) in cases (coloured) and controls (grey). Each dot is one ECG; the half-violin shows the density, the bar the IQR and the circle the median. Hedges' g with 95% CI; p for the primary endpoint and BH-adjusted q for secondary markers.

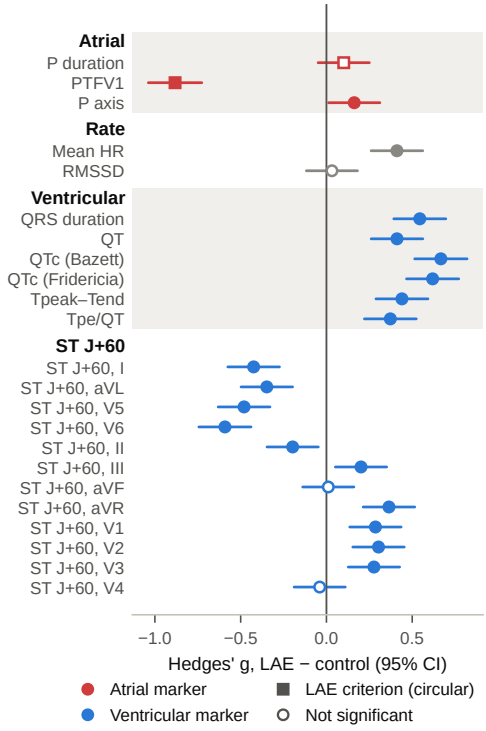


Fig. 6. Adjusted effect size of every marker. Squares: markers that are part of the LAE definition. Open symbols: not significant. The number of beats, a quality covariate counted among the 23 secondary markers, is not plotted.

reports confirmed by a cardiologist (188 cases, 374 controls) it was +1.8 ms (−1.6 to +5.3; $p = 0.29$). In Design B, where the case label (AF in another ECG) does not depend on the

TABLE IV
POST-HOC CHECK FOR LVH AND CONDUCTION DISEASE: ADJUSTED CASE–CONTROL DIFFERENCE (95% CI)

Marker	Original 259/516	(a) excluded 148/438	(b) LVH covariate 259/516
P dur. (ms)	+2.0 (−0.9, +5.0)	+3.9 (+0.4, +7.5)	+1.6 (−1.4, +4.6)
PTFV1 (μV·ms)	−3627 (−4237, −3017)	−3430 (−4153, −2706)	−3351 (−3974, −2728)
QRS (ms)	+14.2 (+10.4, +17.9)	+9.5 (+5.6, +13.4)	+16.6 (+12.8, +20.4)
QTcF (ms)	+32.8 (+24.9, +40.6)	+32.4 (+22.8, +42.0)	+33.6 (+25.6, +41.7)
HR (bpm)	+4.9 (+3.2, +6.7)	+3.6 (+1.4, +5.8)	+5.0 (+3.2, +6.8)

(a) ECGs with LVH, complete bundle branch block, WPW or pacing removed from both groups, not re-matched. Model: marker ∼ group + age + sex (+ LVH in b).

P wave, P duration was longer in cases by +5.2 ms (+0.2 to +10.3; $p = 0.04$; $g = +0.27$; 85 cases, 191 controls); as noted above, this is an association with AF history.

LVH was more common in cases than controls (25.9% vs. 8.5%), as were left (10.0% vs. 3.3%) and right (7.3% vs. 3.3%) bundle branch block. Two post-hoc checks (Table IV) refitted the model (a) after removing every ECG with LVH, complete bundle branch block, WPW or pacing (111 cases and 78 controls) and (b) with LVH as an extra covariate. PTFV1, QRS, QTc and HR kept their direction and significance in both. The QRS difference fell from +14.2 to +9.5 ms in (a), so part of it came from these conditions. P duration stayed non-significant with LVH as a covariate; in the unmatched subset (a) it reached +3.9 ms ($p = 0.03$, uncorrected), which we do not treat as a finding.

F. Classifiers

Table V and Fig. 7 give the results. With all markers (set A), the random forest reached an AUROC of 0.830 (0.800–0.858); logistic regression was close behind (0.810). Without any P-

TABLE V
CLASSIFIER RESULTS, NESTED CROSS-VALIDATION, LAE VS. CONTROL ($n = 775$, PREVALENCE 0.334)

Set	Model	AUROC (95% CI)	AUPRC	Sens	Spec
A	Logistic	0.810 (0.777–0.841)	0.699	0.683	0.781
	Tree	0.750 (0.712–0.785)	0.556	0.710	0.705
	Forest	**0.830** (0.800–0.858)	0.680	0.776	0.717
	Boosting	0.825 (0.792–0.855)	0.695	0.749	0.746
B	Logistic	0.723 (0.684–0.760)	0.580	0.606	0.719
	Tree	0.693 (0.654–0.730)	0.519	0.595	0.736
	Forest	**0.746** (0.708–0.781)	0.593	0.672	0.698
	Boosting	0.734 (0.693–0.771)	0.603	0.676	0.690
	Always "control"	0.500	0.334	0.000	1.000

Set A: all markers plus age and sex. Set B: set A without any P-wave marker. Sensitivity and specificity at the threshold chosen in the inner loop.

wave marker (set B), the best AUROC was 0.746 (0.708–0.781). None of the 200 label shuffles reached the observed AUROC in either set (highest 0.61; $p = 0.005$, the smallest value possible with 200 shuffles). At the threshold chosen in the inner loop, the set-A forest had a sensitivity of 0.776 and a specificity of 0.717 (true positives 201, false negatives 58, false positives 146, true negatives 370). Its accuracy of 0.737 should be read against 0.666 for a model that always answers "control"; this is why accuracy is not the main measure. PTFV1 was the most important marker in set A; in set B the leading markers were ST in V6, QRS duration and QTc (Fig. 8). Because QT, QTcB, QTcF and Tpe/QT are strongly correlated, shuffling them one at a time understates them. Shuffled together (post hoc), the QT family lowered AUROC by 0.029 in set A and 0.046 in set B, against 0.016 and 0.019 for the sum of its single-marker values; the P-wave family lowered it by 0.163 in set A, almost all from PTFV1.

V. Discussion

Main findings. PULSE-AI v1 measures named ECG markers automatically and reports how accurate the measurement is. In an older hospital cohort these markers separate expert-labelled LAE from matched controls, but not through P duration, the marker usually mentioned first. P duration did not differ, and the 120 ms IAB threshold labelled more than nine in ten ECGs as abnormal in *both* groups. PTFV1 separated the groups.

Why P duration did not separate the groups. We see three reasons. First, normal values depend on how P duration is measured: measuring across all leads gives longer values [13], and with such methods healthy men over 60 already have a median near 120 ms [14]. A 120 ms limit taken from visual reading [5] therefore cannot separate groups whose mean age is 68. Second, our measurement has a positive offset (18–28 ms), a slope below one (0.782) and a per-record error (SD 23.4 ms) that is large compared with the expected group difference; all three push the result towards no difference. Third, the ECG criteria for LAE include the negative end of the P wave in V1 [6], which is exactly what PTFV1 measures. In Design B, where the label does not come from the P wave, P duration was longer in cases (+5.2 ms). This suggests that P duration does follow atrial disease, but it should be used as a calibrated continuous value and not as a yes/no call at 120 ms.

The ventricular pattern. A wider QRS, longer QTc, ST depression in the lateral leads and ST elevation in aVR together form the ECG pattern of left ventricular hypertrophy with strain, which often comes with high blood pressure and an overloaded left atrium. These markers are not part of the LAE definition, yet on their own they gave an AUROC of 0.746. LVH was indeed three times as common in cases, and removing it reduced the QRS difference but did not remove it. The classifier is therefore not simply copying the label; the LAE label in PTB-XL marks a wider heart condition.

Treatment. The ARCADIA trial gave apixaban or aspirin to patients with cryptogenic stroke and atrial cardiopathy, defined partly by PTFV1, and was stopped early because the two arms did equally well (HR 1.00, 95% CI 0.64–1.55) [26]. ARCADIA tested a treatment, not a marker. This paper makes no treatment claim. Its contribution is the measurement, which any future study that selects patients by these markers will need.

Limitations. LAE labels in PTB-XL are partly based on the P wave, so atrial differences confirm the measurement rather than reveal new disease. The data are cross-sectional and contain no outcomes; Design B mixes ECGs recorded before and after AF. P onset and QRS onset exceed the CSE limits, and our V1 P offset is late, so PTFV1 threshold calls over-report. The quality check removed 21.0% of ECGs at the same rate in both groups, but for different reasons: controls more often under the QRS-onset rule, cases more often under the QRS-range rule. Ectopic beats were not removed, LVH and conduction disease were handled only by post-hoc checks, and QTc used one threshold for both sexes. Ten-second ECGs cannot capture AF that comes and goes, or long-window heart-rate variability. The classifier analysis was added after the plan, the permutation p is not corrected for choosing the best of four models, and no external cohort was available.

Next steps. The same code runs on single-lead recordings and skips the markers that need missing leads. Our wearable prototype (Fig. 9) records single-lead ECG, photoplethysmography (PPG) and EEG continuously. Long recordings allow RR-interval screening for AF that comes and goes, and valid long-window heart-rate variability; PPG allows blood-pressure trends to be studied. A single lead cannot give every atrial marker: P axis and IAB need the limb leads and PTFV1 needs the chest lead V1. Rhythm, QRS and QTc need only one lead, and in set B these carried most of the signal. Measuring which markers keep their accuracy on a single lead, first on PTB-XL and then on the device against a simultaneous 12-lead ECG, is the aim of PULSE-AI v2.

VI. Conclusion

PULSE-AI v1 measured named atrial and ventricular ECG markers with stated error. In 775 PTB-XL ECGs, P duration with the usual 120 ms threshold did not separate expert-labelled LAE from matched controls, while PTFV1 and ventricular markers did (AUROC 0.830; 0.746 without P-wave

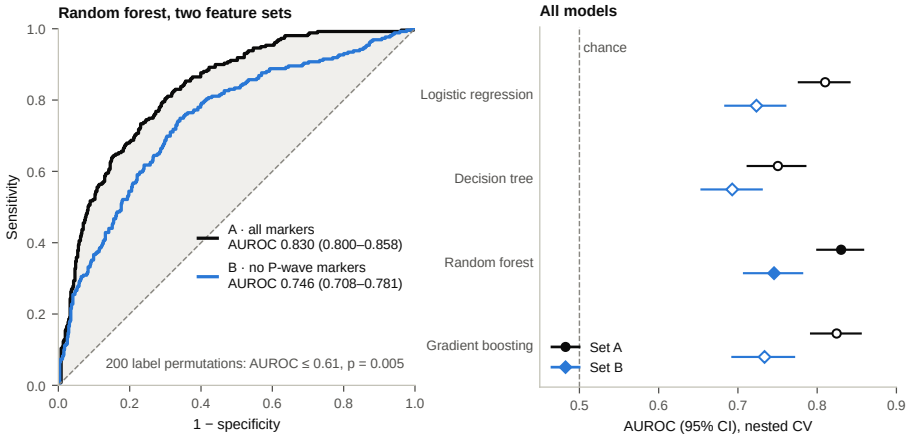


Fig. 7. Left: ROC curves of the random forest with and without P-wave markers (pooled outer folds). Right: AUROC with 95% CI for all four models.

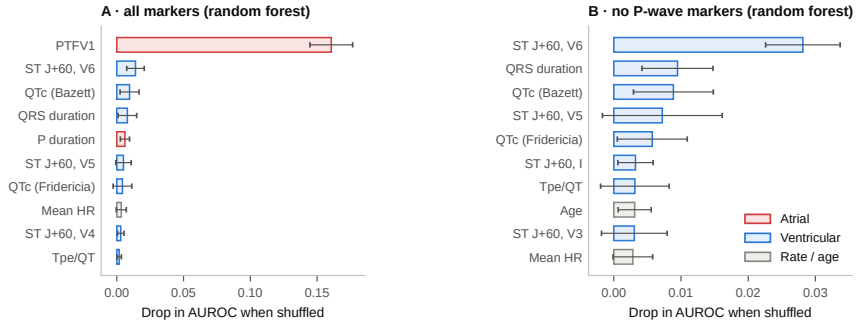


Fig. 8. Permutation importance (drop in outer-fold AUROC when a marker is shuffled), random forest. Error bars: standard deviation over outer folds.

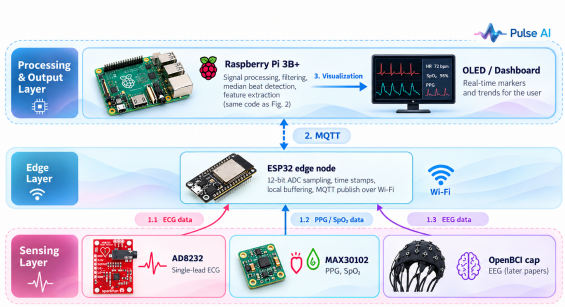


Fig. 9. PULSE-AI wearable prototype in three layers. Numbered arrows give the data path. Illustration only: the dashboard values shown are not measurements, and no result in this paper uses device data.

markers). Automated P-wave analysis should report calibrated continuous values rather than thresholds taken from single-lead reading.

Declarations

Ethics. Only public, de-identified databases were used (PTB-XL, LUDB and QTDB, via PhysioNet [27]). No new data were collected and ethics approval was not required. *Data availability.* All data are openly available on PhysioNet; code is available at the repository given in Section III-H. *Funding.* This work was supported by Dawood University of Engineering and Technology and by the authors' own funds. *Use of AI tools.* AI-based tools were used to assist with code development, language editing and preparation of Figs. 1, 2, 3 and 9. The authors checked all content, analyses and references and take full responsibility for them. *Competing interests.* The authors declare no competing interests.

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